Sample: TCP Connection Basics (lecture notes)

1. Three-way handshake
A TCP connection opens with three segments. The client sends SYN with an initial sequence number. The server replies SYN-ACK, acknowledging the client's number and sending its own. The client finishes with ACK.
Only after this exchange can either side send data.

2. Why three segments and not two?
Both sides must confirm that the other received their initial sequence number. With two segments the server could never be sure its SYN arrived, and an old duplicate SYN could open a stale connection.

3. Connection teardown
Closing uses FIN and ACK in each direction, usually four segments. The side that closes first waits in TIME_WAIT for 2 x MSL so that delayed segments from the old connection die out before the port is reused.

Flow control vs congestion control

4. Flow control
The receiver advertises a window (rwnd) in every ACK: how many more bytes it can buffer. The sender never has more unacknowledged data in flight than rwnd. This protects a slow receiver.

5. Congestion control
The sender also keeps a congestion window (cwnd). Slow start doubles cwnd every round trip until it reaches ssthresh, then congestion avoidance adds one segment per round trip. On a timeout cwnd falls back to 1 segment; on three duplicate ACKs (fast retransmit) it is halved.

6. Key numbers to remember
Effective window = min(rwnd, cwnd).
Default MSS on Ethernet is 1460 bytes (1500 MTU - 40 bytes of headers).

Try asking: "Why does TCP need three segments to open a connection?" or "What happens to cwnd after three duplicate ACKs?"